Synchronous Remote Sensing Boundary Layer Temperature and Humidity Profiles From Ground-Based Infrared Hyperspectral Radiometer Based on Radial Basis Function Neural Networks

Yao Xiao, Shuai Hu, Ruijun Dang, Wanxia Deng, Youfu Li, Wei Huang, Lei Liu, and Wanying Yang

***Abstract*—The accurate and efficient retrieval of boundary layer temperature and humidity profiles is crucial for the study of weather forecasting and vertical transport of pollutants and energy. Aiming at this problem, this article proposes an integrated inversion algorithm to obtain the boundary layer temperature and humidity profiles from ground-based infrared hyperspectral radiation by using radial basis function neural networks (RBFNNs). By using this method, the nonlinear atmospheric radiative transfer processes can be perceived from observational data directly, and the temperature and humidity profiles can be simultaneously inverted without iterative solutions. To investigate the performance of RBFNN, a systematic comparison with the eigenvector method and physical retrieval method is conducted, and the performance of joint and nonjoint retrieval strategies for temperature and humidity is also analyzed. The results indicate that: 1) RBFNN outperforms traditional methods (e.g., eigenvector method and physical retrieval method) on the whole, with root-mean-squared error (RMSE) of 0.8110 K and 1.0070 g/kg for temperature and humidity profiles, respectively; 2) RBFNN demonstrates robust performance in the vertical direction, with near-surface temperature deviations controlled within ± 1 K and humidity deviations maintained below ± 1 g/kg throughout the entire layer; and 3) the combination of temperature and humidity channel in RBFNN enhances retrieval accuracy, with the correlation coefficients (R^2) for temperature and humidity inversions reaching 0.9913 and 0.9461, respectively, outperforming the nonjoint inversions. It is confirmed that RBFNN can effectively improve the retrieval accuracy and stability of the temperature and humidity profiles, providing an efficient and reliable approach for ground-based remote sensing of the boundary layer.**

***Index Terms*—Atmospheric profiles, boundary layer, radial basis function neural network (RBFNN), remote sensing.**

Received 29 January 2026; revised 20 March 2026; accepted 8 April 2026. Date of publication 16 April 2026; date of current version 13 May 2026. This work was supported in part by the Science Foundation of Hunan Province under Grant 2024JJ2058; in part by Hunan Provincial Meteorological Bureau 2026 Innovation and Development Special Project under Grant CXFZ2026-MSXM38; and in part by the National Science Foundation of China under Grant 42175154, Grant 42205122, Grant 62105367, and Grant 62306323. *(Corresponding author: Shuai Hu.)*

Yao Xiao is with the College of Meteorology and Oceanography, National University of Defense Technology, Changsha 410003, China, also with the Key Laboratory of High Impact Weather (Special), China Meteorological Administration, Changsha 410003, China, also with the Meteorological Administration, Yueyang 414000, China, and also with Dongting Lake National Climatic Observatory, Yueyang 414000, China (e-mail: xiaoyao_1024@163.com).

Shuai Hu, Wanxia Deng, and Lei Liu are with the College of Meteorology and Oceanography, National University of Defense Technology, Changsha 410003, China, and also with the Key Laboratory of High Impact Weather (Special), China Meteorological Administration, Changsha 410003, China (e-mail: hushuai2012@nudt.edu.cn; dengwanxia14@nudt.edu.cn; liulei17c@nudt.edu.cn).

Ruijun Dang and Youfu Li are with the College of Meteorology and Oceanography, National University of Defense Technology, Changsha 410003, China (e-mail: dangruijun20@nudt.edu.cn; liyoufu21@nudt.edu.cn).

Wei Huang is with the State Key Laboratory of Complex Electromagnetic Environment Effects on Electronic and Information Systems, Luoyang 471003, China (e-mail: whuang edu@outlook.com).

Wanying Yang is with Suizhou Meteorological Bureau, Suizhou 441300, China (e-mail: yangwanying20@alumni.nudt.edu.cn).

Digital Object Identifier 10.1109/TGRS.2026.3684746

I. Introduction

As an important parameter to describe the thermal dynamic characteristics of the boundary layer [1], the temperature and humidity profiles are directly related to the stability of the atmosphere, the diffusion of atmospheric pollutants, and the evolution of weather systems. Due to these reasons, it plays an indispensable role in meteorological forecasting and environmental science research [2], [3], and obtaining temperature and humidity profiles accurately and efficiently has always been an important prerequisite in the study of the thermal and dynamic properties of the atmosphere.

At present, radiosondes are still the basic means to obtain the vertical distribution of the temperature and humidity of the boundary layer [4], but their application is limited by the high costs and low time resolution, making it difficult to fully capture the daily variation of the atmospheric vertical structure [5]. In contrast, passive satellite remote sensing technology provides another way to acquire atmospheric profiles [6]. However, signals received by satellites are a mixture of the emission of the Earth's surface and atmosphere, making it difficult to effectively invert the temperature and humidity profiles in the near-ground region [7]. Massaro et al. [5] research shows that passive satellite remote sensing will suffer from loss of precision in the retrieval of the atmospheric profiles in the lower atmosphere. Ground-based remote sensing is an important supplement to satellite observation because it is not affected by the ground and can realize continuous observation. Common ground-based remote sensing methods include ground-based microwave radiometers (MWRs), Raman lidar

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:53:08 UTC from IEEE Xplore. Restrictions apply.

(RL), and infrared hyperspectral radiometers. MWR cannot only detect the temperature and humidity profiles but also reveal the microphysical information of the clouds due to its long wavelength [8]. However, there are fewer absorption lines in microwave bands, and the information content in the observation data of MWR is relatively low. Due to this reason, its retrieval accuracy in the boundary layer is limited [9]. RL performs well in detecting tropospheric water vapor profiles and aerosols, but is severely affected by sunlight interference, limiting its effective application during the daytime [10]. In recent years, the rapid development of the Fourier transform infrared hyperspectral measurement technique has enabled the simultaneous observation of the radiance in thousands of spectral channels, thereby more precisely and accurately retrieving the vertical profiles of atmospheric temperature and humidity [11]. Currently, the Atmospheric Emitted Radiance Interferometer (AERI) developed by the University of Wisconsin [12] can provide high-quality information on the lower tropospheric atmospheric profiles [13], [14] and has been widely deployed in the atmospheric radiation measurement (ARM) program.

To ensure the retrieval accuracy of the temperature and humidity profiles, it is crucial to design a reasonable retrieval algorithm. In the early infrared hyperspectral profile inversions, the eigenvector method is generally used [15], [16], but this method is essentially established based on multiple linear regression and does not fully consider the nonlinear characteristics of the radiative transfer process, limiting the improvement of retrieval accuracy [17]. Currently, the physical retrieval methods are widely used in the ground-based infrared hyperspectral atmospheric profile retrieval. The "onion peeling" algorithm [18] is an early version of the physical retrieval method. Due to the ill-posed nature of the retrieval problem, it is highly dependent on initial values, especially the accuracy of the lower atmospheric layer [19]. To address this limitation, Turner and Löhnert [19] proposed the AERIoe retrieval algorithm. This algorithm is established based on Bayesian estimation principles and employs a Gaussian–Newton iteration to obtain convergent results; studies show that it can significantly improve the accuracy and stability of the retrieval process. Wei et al. [20] introduced the L-curve method to optimize the selection of regularization operators for the AERIoe retrieval algorithm, further improving the retrieval efficiency of the algorithm. On the whole, the physical retrieval methods involve complex radiative transfer calculations [21], [22], resulting in large computational loads and long-time consumption. Therefore, statistical retrieval methods are gradually gaining attention as an important means of improving computational efficiency [23].

With the development of deep learning technologies, such models as D3Former [24], HyperSIGMA [25], and Prithvi [26] are established and widely applied in the field of remote sensing. Under this trend, neural network methods have also been increasingly applied in the retrieval of atmospheric temperature and humidity profiles [27], [28], [29], [30], [31]. Compared with physical retrieval algorithms, neural network methods do not require in-depth consideration of complex radiation transfer processes, resulting in faster inversion efficiency. At the same time, with its powerful feature extraction and generalization abilities, neural networks can establish the nonlinear relationship between atmospheric profiles and radiation, which can realize the inversion of temperature and humidity accurately [32]. Yan et al. [33] developed a retrieval model based on batch normalization and robust neural network (BRNN) for MWRs, which has shown that the retrieval accuracy of temperature and humidity profiles can be improved. Cai et al. [31] used an artificial neural network (ANN) to achieve atmospheric profile retrieval for FY-4A/GIIRS, and the retrieval results were consistent with the products of AIRS. In recent years, deep learning technology has opened up new opportunities for the development of remote sensing. Yang et al. [34] proposed an enhanced multiscale feature fusion network that integrates features across different scales, improving the robustness of hyperspectral image classification. To overcome limited sample availability, Fu et al. [35] developed a convolutional neural network (CNN)–Transformer and channel–spatial attention-based network for hyperspectral image classification with few samples. Zhang et al. [36] further demonstrated the transferability of such hybrid architectures by applying a similar multiscale CNN–Transformer to super-resolution tasks. These successful cases demonstrate the potential of neural network methods in atmospheric remote sensing retrieval.

Among numerous neural network models, the radial basis function neural network (RBFNN) is a high-performance model that was first proposed by Broomhead and Lowe [37]. Due to its general approximation capabilities and simple structure [38], RBFNN has been successfully applied in fields such as nonlinear modeling [39], intelligent control [40], and pattern recognition [41]. Compared with traditional neural networks, RBFNN has significant advantages in classification, regression, and function approximation [42], [43]. Based on these advantages, an RBFNN-based inversion framework is proposed for the retrieval of atmospheric temperature and humidity profiles. Unlike conventional physical retrieval methods that rely on iterative radiative transfer calculations, the proposed approach directly learns the nonlinear mapping from infrared observations to atmospheric profiles and can avoid the ill-posed problem in the iteration. By leveraging RBFNN's strong local approximation capability, this framework cannot only achieve higher inversion accuracy of the boundary layer profiles but also significantly reduce the computational complexity, thereby offering a more efficient alternative to both physical and existing neural network methods.

At the same time, atmospheric temperature and humidity exhibit strong coupling effects during atmospheric radiative transfer. On one hand, the temperature determines the ability of water vapor to emit infrared radiation according to the blackbody radiation law. On the other hand, temperature directly regulates humidity distribution by influencing the saturated vapor pressure of air, while humidity in turn can affect the thermodynamic structure of the atmosphere, which is governed by certain atmospheric physical laws, such as the thermodynamics of moist air. To take the inherent connections

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:53:08 UTC from IEEE Xplore. Restrictions apply.



Fig. 1. Specific geographic location of the SGP site.

TABLE I
CHANNEL USED FOR THE RETRIEVAL OF TEMPERATURE AND HUMIDITY PROFILES

Temperature	Humidity
612-618 cm^{-1}	538-588 cm^{-1}
624-660 cm^{-1}	1250-1350 cm^{-1}
674-713 cm^{-1}	
2223-2260 cm^{-1}	

between them into the retrieval process, this article also attempts to combine the temperature and humidity channels in the inversion process and construct an integrated atmospheric inversion model based on RBFNN, and explore the potential of RBFNN in joint inversion of atmospheric temperature and humidity profiles synchronously. This is the second motivation of this work.

The structure of this article is organized as follows. Section II mainly introduces the data used and the design of the RBFNN retrieval model. In Section III, the performance of this method is investigated through actual retrieval experiments, and the performance of joint and nonjoint retrieval strategies for temperature and humidity is also analyzed. Section IV is a brief summary of this article. The purpose of this work lies in the following two aspects. First, by considering the strong coupling effects between atmospheric temperature and humidity in atmospheric physical processes in RBFNN, we try to achieve higher retrieval accuracy of temperature and humidity profiles. Second, in terms of methodology, by introducing RBFNN into the integrated retrieval of temperature and humidity profiles, we try to improve both retrieval accuracy and efficiency compared with traditional inversion methods.

II. DATA

The dataset used in this article was obtained from observation stations located in the South Great Plain (SGP, at 36°36′ N and 97°29′ W, see Fig. 1) and covers the period from 2006 to 2024. It includes radiation data observed by the AERI and the atmospheric profiles obtained by the radiosonde. Given that AERI has multiple observation channels, its data not only covers temperature and humidity inversion channels but also includes trace gas information such as O_3 and CH_4. Therefore, when retrieving temperature and humidity profiles, it is necessary to carefully select appropriate channels to ensure data quality [44]. Specifically, temperature profile retrieval typically relies on sensitive wavelengths of stable gas components (such as CO_2) [18], while humidity profile retrieval focuses on water vapor-sensitive channels. In this process, a total of 244 channels were used for temperature profile retrieval, while 311 channels were used for humidity profile retrieval, which are shown in Table I.

To construct the dataset that contains the infrared hyperspectral radiometer (AERI) and atmospheric profiles, we performed a rigorous preprocessing workflow on the radiosonde data and AERI radiation data from 2006 to 2024, including spatial–temporal matching, quality control, and cloud removal. During the time matching process, the launch time of the radiosondes is taken as the baseline, and the infrared hyperspectral data are matched by selecting the time nearest to the release time of radiosondes (generally, smaller than 8 min). For the spatial matching, both the radiosonde and the infrared hyperspectral radiometer operate at the same site, and their distance is smaller than 150 m.

After selecting, a total of 13 764 data samples that met the predefined criteria were obtained. For further analysis, the dataset was divided into a training dataset and a validation dataset. The training dataset comprises 12 160 datasets from 2006 to 2022, intended for model training and optimization, and the validation dataset includes 1604 datasets from 2023 to 2024, used to evaluate model performance. Each dataset includes temperature and humidity data obtained by AERI and the radiosonde at all retrieval height layers below 3 km (a total of 37 layers).

III. METHODS

RBFNN is a classic technique for multidimensional space interpolation [45], [46]. Its design is based on function approximation theory and belongs to the category of feedforward networks. Compared with BP and other neural networks, it has better nonlinear fitting accuracy, recognition ability, and learning efficiency, and it is not easy to fall into local optimization as well. Therefore, the RBFNN neural network is widely used in the fields of pattern recognition and state monitoring. Due to its unique advantages, a joint inversion model for atmospheric temperature and humidity profiles based on RBFFN was designed in this article, its retrieval framework is presented in Fig. 2. The process of RBFNN retrieval model can be viewed as finding the optimal fitting plane in a multidimensional space to match the training data, where the input parameters are the infrared radiances, and the output parameters are set as the temperature and humidity profiles.

Since temperature and humidity are governed by certain atmospheric physical laws (such as the thermodynamics of moist air and dynamical principles), they imply some inherent connections between them. By jointly learning and inverting between the infrared radiance and the two parameters, it is possible to incorporate the implicit constraints of these atmospheric laws into the training process.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:53:08 UTC from IEEE Xplore. Restrictions apply.

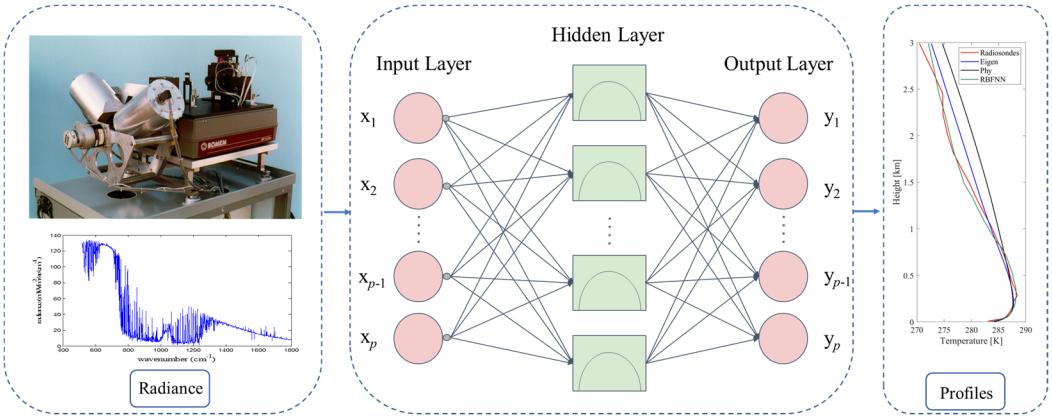


Fig. 2. RBFNN retrieval framework.

In the RBFNN, the observed radiance of the infrared hyperspectral radiometer AERI is $\mathrm{x}^a = \{\mathrm{x}_i^a\}_{i=1}^N$, where x_i^a represents the ith sample of the observed radiances and N is the total number of samples obtained. The corresponding true values are obtained from the radiosonde and are presented in $\mathrm{Y} = \{\mathrm{y}_i\}_{i=1}^N$, where y_i represents the measured temperature and humidity measured by the radiosonde at the same time, written as $\mathrm{y}_i = [\mathrm{y}_i^t, \mathrm{y}_i^q]$. Here, y_i^t and y_i^q are the temperature and humidity vectors, respectively. y_i^t and y_i^q are expressed as $\mathrm{y}_i^t = [t_1, t_2, \ldots t_m]^T$ and $\mathrm{y}_i^q = [q_1, q_2, \ldots q_m]^T$, respectively. t_j and q_j represent the temperature and water vapor mixing ratio at the jth vertical, respectively. m represents the total number of vertical heights. "T" represents the transpose operation of the matrix. In the retrieval process, m is set as 37, namely, the atmospheric layer is divided into 37 sublayers.

In this retrieval scheme, a three-layer structure is adopted by the RBFNN [38], including an input layer, a hidden layer, and an output layer, as shown in Fig. 2. The input layer is first mapped to a higher dimensional hidden layer, and the hidden layer and the output layer are connected by weighting coefficients [47].

Unlike a traditional feedforward neural network (FFNN), RBFNN introduces a nonlinear transformation between the input layer and the hidden layer. The hidden layer performs a nonlinear transformation on the input parameters, and the output layer acts as a linear combiner that maps the nonlinear transformation to a new space. In the hidden layers, the radial basis functions $\varphi(\mathrm{r})$ are used as the activation functions, and it determines its recognition and nonlinear fitting ability. Therefore, the selection of the basis function is the key factor in the entire retrieval model. In our retrieval scheme, all nodes use the same Gaussian radial basis function, written as

$$\varphi_i(\mathrm{x}) = \varphi\left(\mathrm{x}_p - \mathrm{c}_i\right) = e^{-\frac{\|\mathrm{x}_p - \mathrm{c}_i\|}{2\sigma_k^2}} \tag{1}$$

where $\varphi(\mathbf{x})$ is a radial basis function and c_i is the center of the hidden layer node. For input infrared radiance x, the output of the RBFNN is given as follows:

$$y_i = \sum_{i=1}^{h} w_{ij}\varphi\left(\mathrm{x}_p - \mathrm{c}_i\right), j = 1, 2, \ldots\ldots \tag{2}$$

x_p denotes the pth input sample, i.e., the radiance observed by the 555 channels of AERI (including 244 temperature-sensitive channels and 311 water vapor-sensitive channels). w_ij denotes the weights of each neuron node from the hidden layer to the output layer. $i = 1, 2, \cdots\cdots$ denotes the number of nodes in the hidden layer. y_i represents the actual output of the jth output node of the neuron network corresponding to the input sample. The input data in this article and the output data are the atmospheric profiles. By using linear optimization methods to minimize the loss function between the output and the true values, radial basis function networks can achieve globally optimal solutions by adjusting the weights w_{ij} and c_i. In the neural network, the mean squared error (mse, the average of the squared differences between the predicted values and the true values) is adopted as the loss function, written as

$$L = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{3}$$

where N is the number of samples, y_i is the true value, and $\hat{y}_i$ is the predicted value.

Referring to the principle of the physical inversion method, we consider the correlation between infrared radiation from different channels and replace the distance in (1), namely, $\|\mathrm{x}_p - \mathrm{c}_i\|$, with the Mahalanobis distance, written as

$$\varphi_i(\mathrm{x}) = \varphi\left(\mathrm{x}_p - \mathrm{c}_i\right) = e^{-\frac{(\mathrm{x}_p - \mathrm{c}_i)^\mathrm{T}\mathrm{S}_e(\mathrm{x}_p - \mathrm{c}_i)}{2\sigma_k^2}} \tag{4}$$

where S_e is the covariance matrix of the infrared radiance, which can be calculated from the dataset established in Section II.

In this retrieval scheme, since the Gaussian function is chosen as the activation function for RBFNN, the diffusion width of the basis function (i.e., σ_k) becomes an important factor affecting network performance (σ determining the "diffusion speed" or "smoothness" of the basis function). In our retrieval scheme, σ is controlled by the spreading parameter, which is the half-width of the radial basis function, written as

$$spread = \sqrt{2\ln 0.5}\cdot\sigma. \tag{5}$$

Generally speaking, as the spread value increases (i.e., as σ becomes larger), the approximation results curve of the RBFNN tends to become smoother. Conversely, as the spread value decreases, the approximation results become steeper.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:53:08 UTC from IEEE Xplore. Restrictions apply.

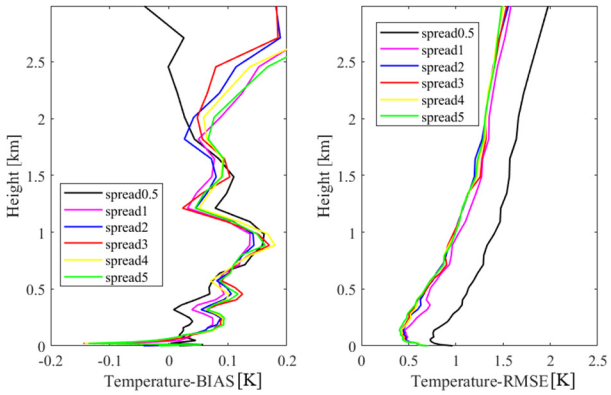


Fig. 3. Vertical distributions of BIAS and RMSE of the retrieved temperature.

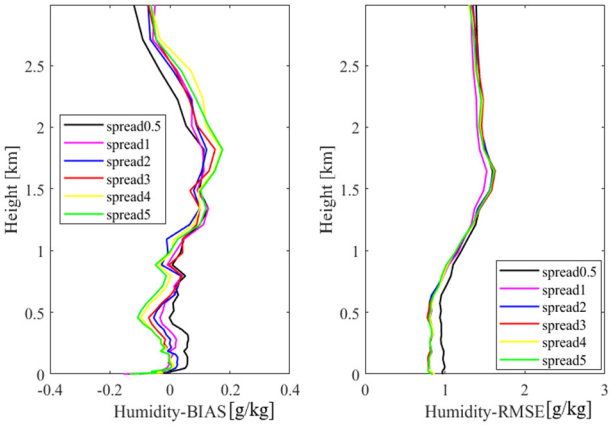


Fig. 4. Vertical distributions of BIAS and RMSE of the retrieved humidity.

TABLE II

BIAS AND RMSE OF THE RETRIEVED TEMPERATURE AND HUMIDITY PROFILES UNDER DIFFERENT SPREAD PARAMETERS

spread	Temperature (K)		Humidity (g/kg)	
	BIAS	RMSE	BIAS	RMSE
0.5	0.055579	1.183085	0.053921	1.12235
1	0.082005	0.858642	0.039542	1.01668
2	0.080225	0.810953	0.039073	1.00698
3	0.084663	0.809172	0.04281	1.01757
4	0.090487	0.803458	0.050562	1.02059
5	0.093267	0.794551	0.055231	1.01338

Therefore, determining the spread value is an important step in the design of the RBFNN retrieval scheme.

Due to the importance of spread value, a series of inversion experiments was conducted to determine a more appropriate spread parameter. In this experiment, the spread parameter varied from 0.5 to 10, the number of nodes (Nn) in the hidden layer was set as 600, and the training dataset described in Section II was utilized to train the RBFNN inversion model. The performance of the retrieval scheme was evaluated using validation set data. The bias (BIAS) and root-mean-squared error (RMSE) of the retrieved temperature and humidity under different spread parameter settings are presented in Table II, while their vertical distributions are illustrated in Figs. 3 and 4, respectively.

From the table, it can be found that when the spreading parameter is 2, the performance of the inversion model is better than in other cases on the whole. Thus, in our RBFNN model, the spreading parameter is set to 2.

The number of nodes in the hidden layer of RBFNN is also an important parameter influencing the fitting ability and generalization performance of the model. To optimize the number of nodes (Nn) in the hidden layer of RBFNN, we have also discussed the relation between the retrieval accuracy and the number of nodes; the results are shown in Table III.

TABLE III

MAE AND RMSE OF THE RETRIEVED TEMPERATURE AND HUMIDITY PROFILES UNDER DIFFERENT NUMBERS OF NODES IN THE HIDDEN LAYER

Nn	Temperature (K)		Humidity (g/kg)	
	MAE	RMSE	MAE	RMSE
500	0.79349	0.87673	1.09679	1.02347
600	0.77254	0.81095	1.07953	1.00668
700	0.77449	0.81737	1.08204	1.00698
800	0.77402	0.81626	1.07955	1.00673
900	0.77863	0.82536	1.08233	1.00636
1000	0.78485	0.82893	1.08279	1.00899
1100	0.79102	0.83186	1.08206	1.01019
1200	0.79235	0.88033	1.07981	1.03435

From the table, it can be seen that the MAE and RMSE of the inversion results first decrease and then increase with the increasing of the number of nodes in the hidden layer, but the variation is not very large. The temperature and humidity profiles both achieve better inversion performance when the number of hidden layer nodes is between 600 and 800. Therefore, in our inversion model, the number of hidden layer nodes is set to 600.

To further verify the robustness of the model, the leave-one-out method is adopted for model verification. In this method, the training dataset was divided into five parts. Then, one part was used as the test dataset sequentially, while the remaining four served as the training dataset to validate the model's inversion results. The experiments show that, in all cases, the correlation coefficients between the retrieved temperature and the measurements from radiosondes were all above 0.94, and the correlation coefficients for the temperature profile are all above 0.98.

IV. RESULTS AND ANALYSIS

To validate the accuracy and effectiveness of RBFNN in retrieving atmospheric profiles, the atmospheric profiles inverted by the RBFNN scheme are compared with the profiles

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:53:08 UTC from IEEE Xplore. Restrictions apply.

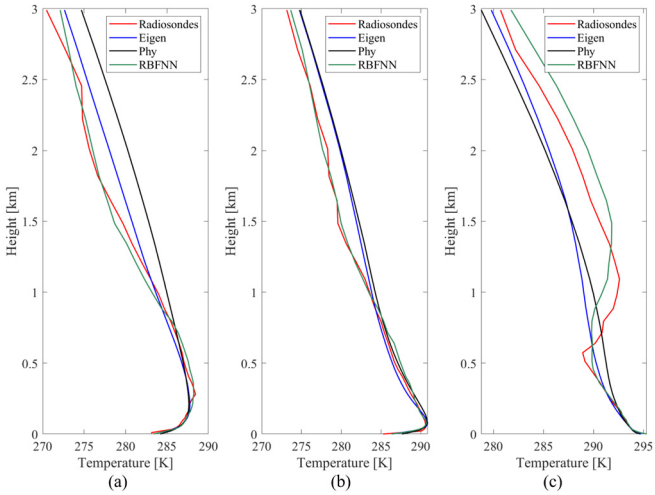


Fig. 5. Three individual cases are selected from (a) 20230221, (b) 20230521, and (c) 20231117 and the performance of RBFNN is compared with the Eigenvector method, and the physical retrieval method, respectively.

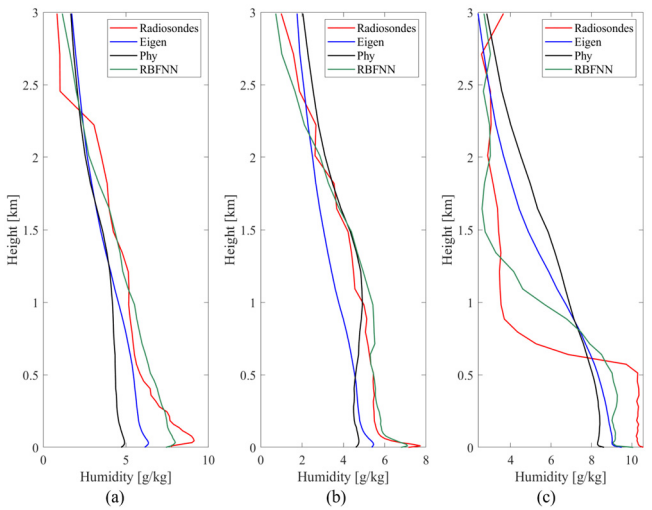


Fig. 6. Humidity profiles obtained by the eigenvector method, physical retrieval method, and RBFNN. Same as Fig. 5, but for humidity profiles. (a) 20230221, (b) 20230521, and (c) 20231117.

obtained by the radiosondes and traditional retrieval methods, such as the eigenvector method and the physical retrieval method. In the eigenvector method, the number of eigenvectors for the temperature retrieval is set as 16, while for the humidity profiles, the number of eigenvectors is set as 20. For the physical retrieval method, the initial value of the iteration is set as the average temperature and humidity profiles in the past ten years. The iteration termination condition is set as

$$\left(\mathrm{x}^{n}-\mathrm{x}^{n+1}\right)^{T}\mathrm{S}\left(\mathrm{x}^{n}-\mathrm{x}^{n+1}\right)\leq N \tag{6}$$

where x^{n} and x^{n+1} are the retrieved atmospheric profiles in the nth and $(n+1)$th iterative steps, respectively; S is the posterior covariance matrix; and N is the dimension of $\mathbf{x}$. For the technical details of the physical inversion technique, the readers can refer to [19], and [48].

The verification process is divided into two parts. The first part is a typical case analysis, and the second part is a statistical analysis. In addition, to further verify that the combination of temperature and humidity channels can enhance retrieval accuracy, the performance of joint and non-joint retrieval strategies for temperature and humidity is also analyzed subsequently.

A. Case Analysis

In the training process of the RBFNN, the atmospheric profiles obtained by the radiosondes are taken as supervisory signals. Then, by gradually optimizing the training parameters of the RBFNN, the output parameters (i.e., the inverted profiles) of the RBFNN approach the temperature and humidity obtained by the radiosondes (the true values). To test the retrieval ability of RBFNN to capture the complex vertical distribution of the atmosphere, some atmospheric profiles with inversion structures were selected for the case study. The three individual cases are selected from 20 230 221, 20 230 521, and 20 231 117, and the performance of RBFNN is compared with the eigenvector method and the physical retrieval method. In these cases, they all exhibited inversion phenomena in the middle and lower atmosphere, and the profiles obtained by the radiosonde were relatively representative, where in the first case, the temperature profile shows a gradual turning point at the altitude of 300 m; in the second case, there is temperature inversion near the Earth surface; in the third case, the temperature profile shows a turning point between 500 m and 1 km in altitude. The comparison of these inversion methods is shown in Figs. 5 and 6.

Fig. 5 shows the comparison of temperature profiles obtained by the eigenvector method, physical retrieval method, RBFNN, and radiosondes. Overall, the temperature profiles obtained by the three retrieval methods accurately depict the variation of atmospheric temperature with altitude and are in good agreement with the radiosonde data. However, in capturing retrieval phenomena, the RBFNN retrieval results demonstrate better performance. For temperature profiles below 500 m, all three algorithms perform well. However, when the altitude exceeds 500 m and temperature profile inflection points occur, both the physical retrieval method and eigenvector method traverse these inflection points in a relatively smooth manner. While the RBFNN exhibits a slightly delayed response to inversion inflection points, its retrieval results are closest to actual radiosonde data, followed by the eigenvector method. The phenomenon indicates that compared with other algorithms, the RBFNN has a stronger ability to extract features of different scales and different levels; therefore, its retrieval accuracy is relatively higher.

Fig. 6 shows a comparison between humidity profiles detected by radiosondes and those obtained by the eigenvector method, physical retrieval method, and RBFNN method with data. It can be seen that all three retrieval methods can capture the fluctuations of humidity in the lower atmosphere. However, the results of RBFNN are closer to the radiosonde data than other algorithms, and it performs better in the turning position of the humidity profiles. The high inversion accuracy demonstrates its powerful nonlinear fitting ability between the infrared radiance and atmospheric humidity. As the altitude

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:53:08 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
AVERAGE RETRIEVAL ERROR OF EIGENVECTOR METHOD (EIGEN), PHYSICAL RETRIEVAL METHOD (PHY), AND RBFNN

Temperature（K）	MAE	RMSE
Phy	2.7068	1.5835
Eigen	1.1912	0.9286
RBFNN	0.7725	0.8110
Humidity（g/kg）	MAE	RMSE
Phy	2.1655	1.4426
Eigen	1.9546	1.3911
RBFNN	1.0795	1.0070

increases, the retrieval accuracy of all these methods decreases because the weighting functions of the ground-based infrared remote sensing are mainly distributed in the lower atmosphere, while the information content in the high altitude is relatively lower. However, in terms of the overall retrieval precision, RBFNN still shows a better inversion effect of water vapor.

B. Statistical Analysis

To further validate the performance of RBFNN, the temperature and humidity profiles in 2023 and 2024 (including 1604 samples in the validation dataset) were inverted, and the mean BIAS and RMSE between the retrieval results of the three methods and the radiosonde data were calculated, respectively. The definitions of BIAS, RMSE, and MAE are written as

$$BIAS\,(i) = \frac{1}{M}\sum_{j=1}^{M}\left(X_{sonde}\,(i,j) - X_{retrieval}\,(i,j)\right) \tag{7}$$

$$RMSE\,(i) = \sqrt{\frac{1}{M}\sum_{j=1}^{M}\left(X_{sonde}\,(i,j) - X_{retrieval}\,(i,j)\right)^2} \tag{8}$$

$$MAE = \frac{1}{M}\sum_{i=1}^{M}\left|X_{retrieval} - X_{sonde}\right| \tag{9}$$

where X_{sonde} represents the temperature or humidity observed by the radiosondes, $X_{retrieval}$ represents the estimated values of temperature or humidity obtained by different retrieval methods, i denotes the sequence number of the altitude layers, j denotes the sequence number of different temperature or humidity samples, and m represents the total number of samples.

Using radiosonde data as the true value, the accuracy of the eigenvector method, physical retrieval method, and RBFNN was evaluated by calculating the RMSE and MAE. The statistical results are summarized in Table IV.

In terms of temperature profile retrieval, RBFNN demonstrated excellent performance, with an RMSE value of 0.8110 K and an MAE value of 0.5910 K. Compared with the eigenvector method, the MAE of RBFNN is 35.1% smaller, and the RMSE is 12.7% smaller. Compared to the physical retrieval method, the MAE of RBFNN was reduced by 71.5% and the RMSE by 48.8%. Similar performance is achieved in the retrieval of the humidity profile, where RMSE and MAE of RBFNN are 1.0070 and 1.0795 g/kg, respectively. Compared with the eigenvector method, RBFNN reduced MAE by 44.8% and RMSE by 27.6%. Compared with the physical retrieval method, RBFNN achieved a reduction of 50% and 30% in MAE and RMSE, respectively. It is worth noting that all three methods performed worse in humidity profile retrieval than in temperature profile retrieval, suggesting that water vapor retrieval is more challenging than temperature retrieval. The reason is that the vertical distribution of water vapor is much more complex than temperature in the atmosphere. From the discussion above, RBFNN produces the smallest RMSE and MAE among the three methods, demonstrating its potential and superiority in the inversion of atmospheric profiles.

Furthermore, the vertical distributions of the BIAS and RMSE of temperature and humidity retrieved by the RBFNN method, the physical retrieval method, and the eigenvector method. The results are shown in Fig. 7, where Fig. 7(a) shows the vertical distribution of BIAS and RMSE of the humidity retrieved by different methods, and Fig. 7(b) shows the results of temperature profiles.

As can be seen from Fig. 7, for the BIAS of humidity profiles below 500 m, the BIAS of the eigenvector method and RBFNN is similar. For the height above 500 m, the BIAS of the physical retrieval method shifts from a negative deviation to a positive deviation and reaches its maximum at an altitude of 1.5 km, while the BIAS of the eigenvector method shifts from a positive deviation to a negative deviation. On the whole, the BIAS of the RBFNN exhibits relatively stable variation with increasing altitude and is closest to the zero line among the three methods. For the RMSE of humidity profiles below 500 m, the RMSE of the eigenvector method and the physical retrieval method generally decreases slightly with increasing altitude, while the RMSE of the RBFNN is relatively small and changes relatively smoothly. Below 1.5 km, the RMSE of all three methods generally increases with altitude, and the deterioration of inversion results is caused by the vertical distribution of the weight function. Within the entire retrieval height layer, the fluctuation of RMSE obtained by the RBFNN is relatively small, indicating that this method has good adaptability and robustness in the retrieval of the humidity profiles. The maximum RMSE values are mostly within 1.5 g/kg, followed by the eigenvector method. From the analysis above, it can be found that the RBFNN inversion scheme achieves the best performance in the retrieval of temperature profiles.

For the BIAS of the temperature inversions, the BIAS of the eigenvector method and RBFNN scheme is basically approaching 0, while that of the physical retrieval method is slightly larger, especially in the region ranging from 500 to 2000 m. Overall, the BIAS of RBFNN tends to be positive values, and its variation curve with the increasing of height is relatively smooth, which indicates that RBFNN might underestimate the temperature in the temperature inversion. From the RMSE, the retrieval accuracy of RBFNN is notably higher than that of the physical retrieval method and the eigenvector retrieval method. For the altitude below 500 m, the RMSE

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:53:08 UTC from IEEE Xplore. Restrictions apply.

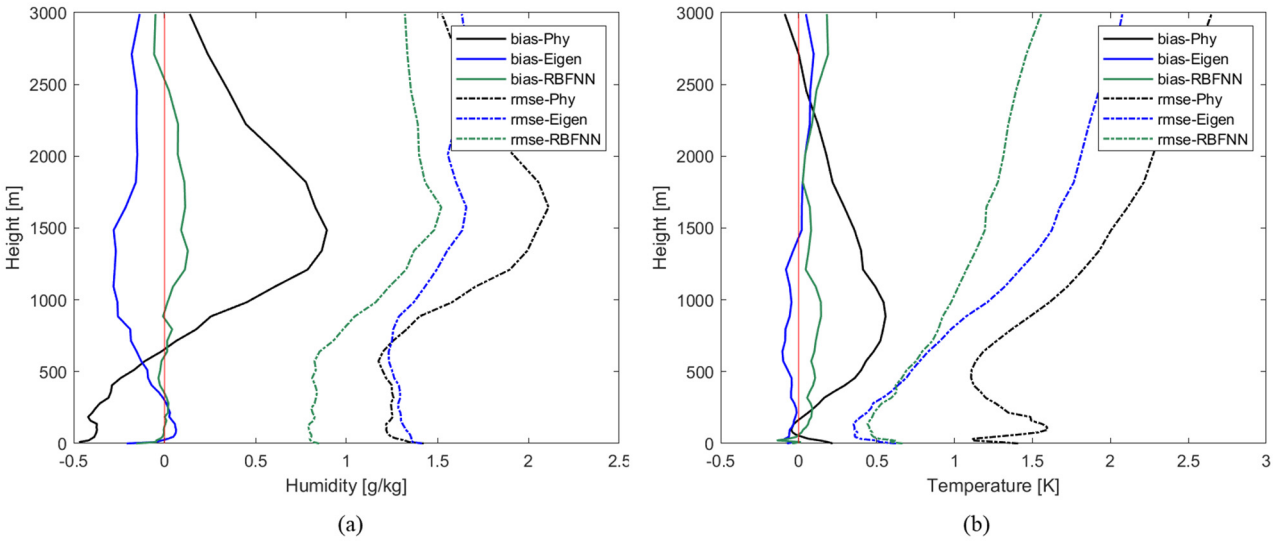


Fig. 7. Vertical distribution of BIAS and RMSE of the eigenvector method, physical retrieval method, and RBFNN for (a) humidity and (b) temperature profiles. The black line represents the physical method results, the blue line represents the eigenvector retrieval method results, and the green line represents the RBFNN results. The colored solid lines represent the BIAS of each method, while the colored dashed lines represent the RMSE. The vertical red line represents zero values.

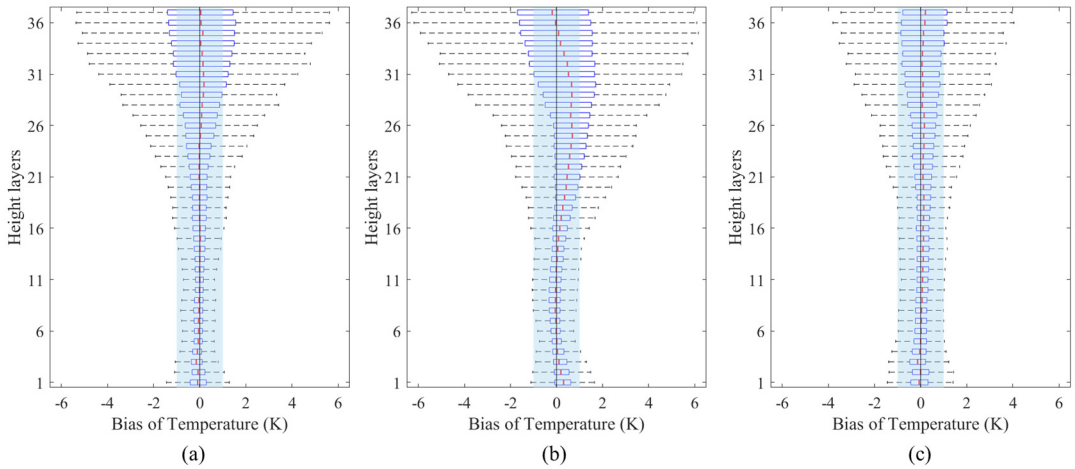


Fig. 8. Statistics of the temperature retrieval errors at different altitudes. (a) Eigenvector method, (b) physical retrieval method, and (c) RBFNN. The short red lines within the boxes represent the median of the errors, the left and right boundaries of the blue box lines represent the first quartile to the third quartile, and the blue shading indicates temperature errors of ± 1 K.

of the RBFNN is similar to that of the eigenvector method, but as the height increases, the RMSE of all three methods increases with height, and the RMSE of the RBFNN remains the smallest. For the altitude below 3 km, the RMSE does not exceed 1.3 g/kg, and within 0–1 km, its RMSE remains below 1 g/kg. Especially, when the altitude is larger than 1 km, the advantage of the RBFNN algorithm becomes more notable, followed by the eigenvector method and physical retrieval method. Within the entire inversion height range, the maximum RMSE values of the RBFNN retrieved temperature profiles are generally within 1.5 K, with minimal fluctuations as altitude changes, indicating high inversion accuracy. This suggests that the RBFNN method has good adaptability and robustness in temperature retrievals.

To further evaluate the performance of the RBFNN inversion algorithm, a statistical analysis of the retrieval errors (both temperature and humidity) is performed at different altitudes as well. The results are shown in Figs. 8 and 9.

As shown in Fig. 8, the temperature deviations predicted by the RBFNN are primarily concentrated within ± 1 K, demonstrating the high retrieval accuracy achieved in the inversion of near-surface temperature profiles. Compared with the eigenvector method and the physical retrieval method, the error intervals of the RBFNN [represented by the length of the boxes, see Fig. 8(a) and (b)] are more compact, meaning that the temperature profiles obtained by this method are more concentrated around the median value. Furthermore, regarding temperature fluctuations at an altitude of 3 km, the average

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:53:08 UTC from IEEE Xplore. Restrictions apply.

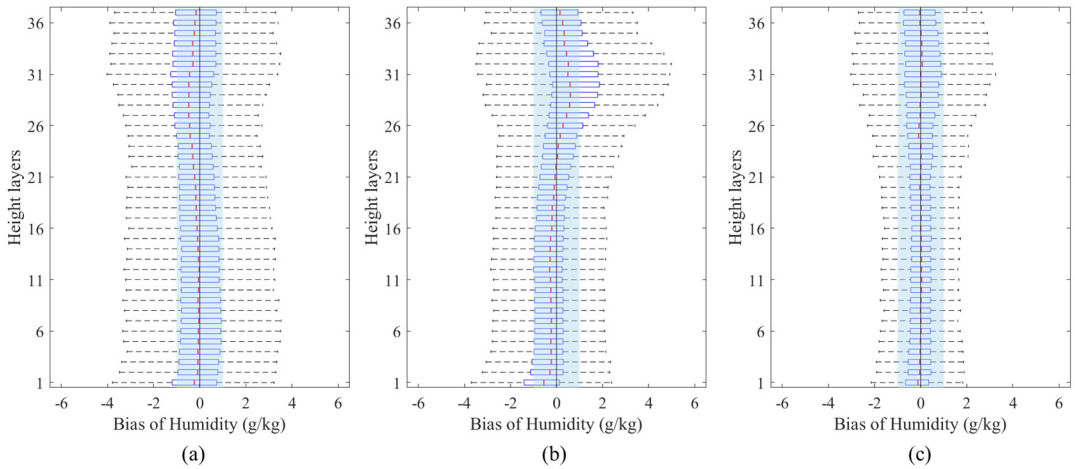


Fig. 9. Statistics of humidity retrieval errors at different altitudes. (a) Eigenvector method, (b) physical retrieval method, and (c) RBFNN. The short red lines within the boxes represent the median of the errors, with the left and right boundaries of the box lines corresponding to the first quartile to the third quartile. The blue shading indicates humidity errors of ±1 g/kg.

temperature deviation obtained by RBFNN is very small in the near-surface region and closer to zero (black line), which further confirms the advantages of RBFNN in the retrieval of temperature profiles.

Fig. 9 shows a comparison of the deviations of humidity profiles inverted by different methods. Specifically, the RBFNN maintains a humidity deviation within ±1 g/kg throughout the entire retrieval layer, and its interquartile range is the smallest among the three methods at any height, demonstrating high accuracy and stability. In contrast, while the humidity profile deviations of the eigenvector method and the physical inversion method also mostly remain within ±1 g/kg, their interquartile ranges are generally wider than those of RBFNN. In addition, in terms of error distribution within 3 km, the average inversion errors of RBFNN are smaller in the near-surface region, closely aligning with the black zero reference line. The statistical analysis further confirms that the RBFNN algorithm can be effectively applied to retrieve the humidity profiles.

C. Comparison of Combined and Noncombined Temperature and Humidity Channels

Since temperature and humidity exhibit strong coupling effects during atmospheric radiative transfer, the temperature and humidity profiles are simultaneously retrieved by the RBFNN inversion model. To evaluate whether the joint inversion scheme can effectively improve the inversion accuracy, the performance of joint and nonjoint retrieval strategies is also analyzed.

Table V presents the MAE and RMSE of temperature and humidity profiles retrieved by RBFNN, where the RBFNN_Y algorithm denotes that the temperature and humidity channels (RBFNN_Y) are combined in the model, while RBFNN_N denotes that the temperature and humidity profiles are retrieved separately. From the results in Table V, it can be found that the RBFNN using the combined temperature and humidity channels demonstrates significant advantages in both temperature and humidity retrieval, validating the effectiveness of the combined inversion strategy.

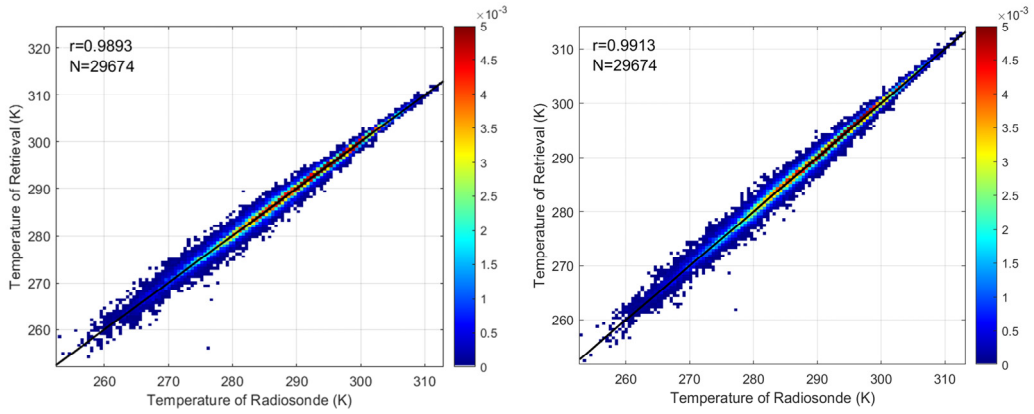
TABLE V
AVERAGE RETRIEVAL ERROR OF RBFNN ("Y" DENOTES THE JOINT RETRIEVAL STRATEGY AND "N" DENOTES THE NONJOINT RETRIEVAL ALGORITHM)

Temperature（K）	MAE	RMSE
RBFNN_Y	0.7725	0.8110
RBFNN_N	0.9520	0.8667
Humidity（g/kg）	MAE	RMSE
RBFNN_Y	1.0795	1.0070
RBFNN_N	1.1669	1.0513

In terms of temperature profile retrieval, the MAE of RBFNN_Y is 0.7725 K, notably lower than 0.9520 K of RBFNN_N, indicating that the combined temperature–humidity channel can effectively reduce the retrieval error of temperature. The same conclusion can be drawn from the RMSEs of the two retrieval strategies, where the temperature RMSE of RBFNN_Y was 0.8110 K, which was approximately 6.4% lower than that of RBFNN_N.

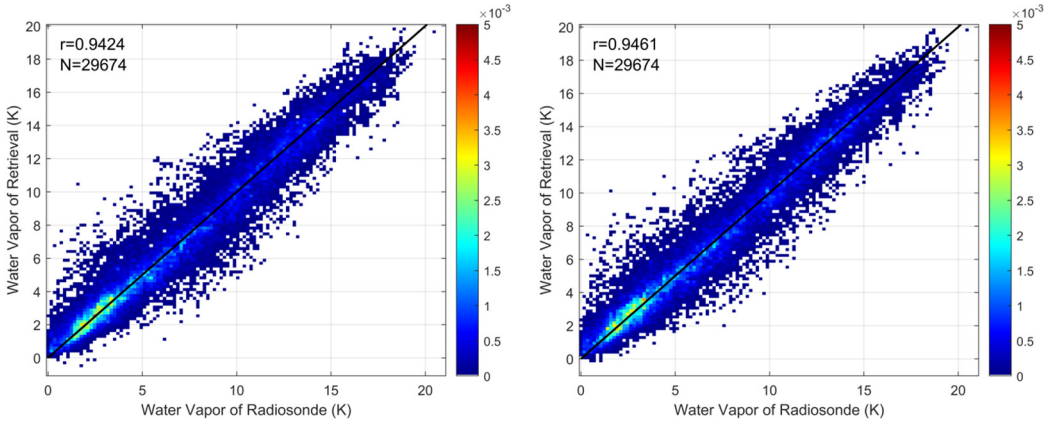
For the retrieval of humidity profiles, the MAE of RBFNN_Y is 1.0795 g/kg, which is lower than the 1.1669 g/kg of RBFNN_N, indicating that the combined temperature–humidity channel also has advantages in humidity retrieval. A similar conclusion can be obtained from the RMSE of humidity, where the RMSE of RBFNN_Y is 1.0070 g/kg, approximately 4.2% lower than that of RBFNN_N (1.0513 g/kg). The inversion algorithm combined temperature–humidity channel (RBFNN_Y) exhibits lower MAE and RMSE in both temperature and humidity inversions, indicating that combined retrieval can more effectively capture the interaction between temperature and humidity, thereby improving retrieval accuracy.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:53:08 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 10. RBFNN [(a) noncombined and (b) combined] retrieval temperature profiles compared with scatter plot of radiosonde data.



(a) (b)

Fig. 11. RBFNN [(a) noncombined and (b) combined] retrieval humidity profiles compared with scatter plot of radiosonde data.

To further investigate the performance of w joint and nonjoint retrieval strategies for temperature and humidity, scatter plots were created using the retrieval results and radiosonde data (1604 samples). Fig. 10 shows the scatter plot comparison between the temperature profiles obtained from RBFNN and radiosonde, while Fig. 11 shows the scatter plot comparison of the humidity profiles. In Figs. 10 and 11, there are two subplots: (a) represents the RBFNN retrieval results with temperature and humidity channels not combined, and (b) represents the RBFNN retrieval results with temperature and humidity channels combined.

In Fig. 10, the correlation coefficient R^2 between the temperature obtained by the joint retrieval strategy and the radiosonde data reaches 0.9913, while the correlation coefficient R^2 for the nonjoint retrieval algorithm is 0.989. For the retrieval of the humidity, the correlation coefficient R^2 of the retrieval algorithm combining temperature and humidity channels reaches 0.9461, while the correlation coefficient R^2 of the noncombined retrieval is only 0.9424. From the discussion above, it can be concluded that joint retrieval strategies have higher inversion accuracy compared to nonjoint ones. This conclusion may be attributed to the fact that temperature and humidity are two closely related and mutually influencing parameters, and their combined retrieval can more accurately capture their interactive relationship, thereby reducing the deviation in the inversion results. Overall, the retrieval accuracy of the RBFNN with a joint retrieval strategy is higher for temperature than for humidity, and the scatter plot of the humidity profile in Fig. 11 is more dispersed than that of the temperature profile in Fig. 10. The phenomenon can be attributed to the complex vertical distribution of water vapor in the atmosphere.

Fig. 12 shows the variation of RMSE of joint and nonjoint retrieval strategies, where the left figure shows the results of temperature, and the right figure shows the results of water vapor. For temperature profiles, though the RMSE of RBFNN_Y and RBFNN_N are similar, RBFNN_Y begins to significantly outperform RBFNN_N above 500 m. For the humidity profile, RBFNN_Y shows a slight advantage over RBFNN_N below 2 km, particularly within the altitude range of 1 ~ 2 km. This analysis further validates that joint retrieval strategies can improve the retrieval of the temperature and humidity profiles to some extent. This phenomenon may be attributed to the fact that temperature and humidity are two closely related and mutually influential parameters, and the joint retrieval strategy can more accurately capture the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:53:08 UTC from IEEE Xplore. Restrictions apply.

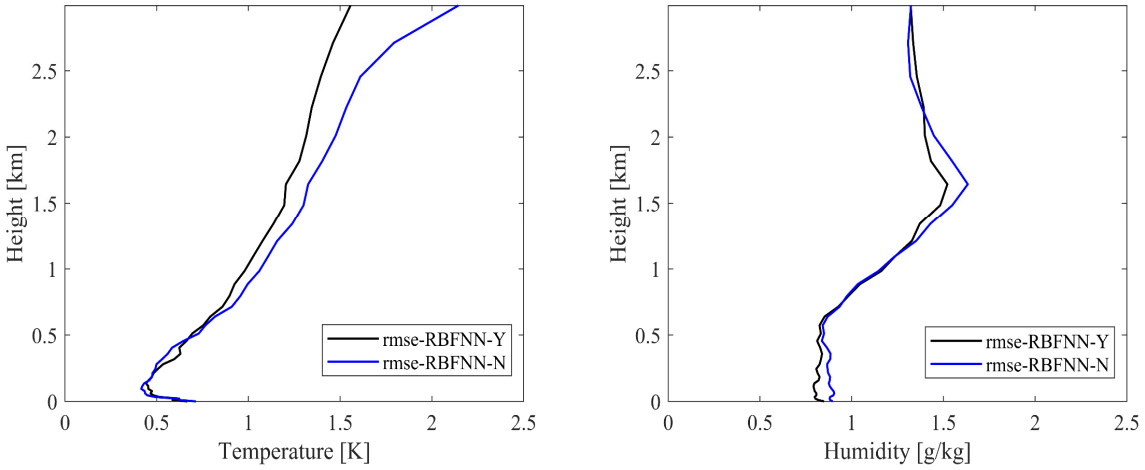


Fig. 12. RMSE of RBFNN (Y denotes combined and N denotes noncombined temperature and humidity channels) as a function of height. The black lines represent the results of RBFNN_Y, and the blue lines represent the results of RBFNN_N.

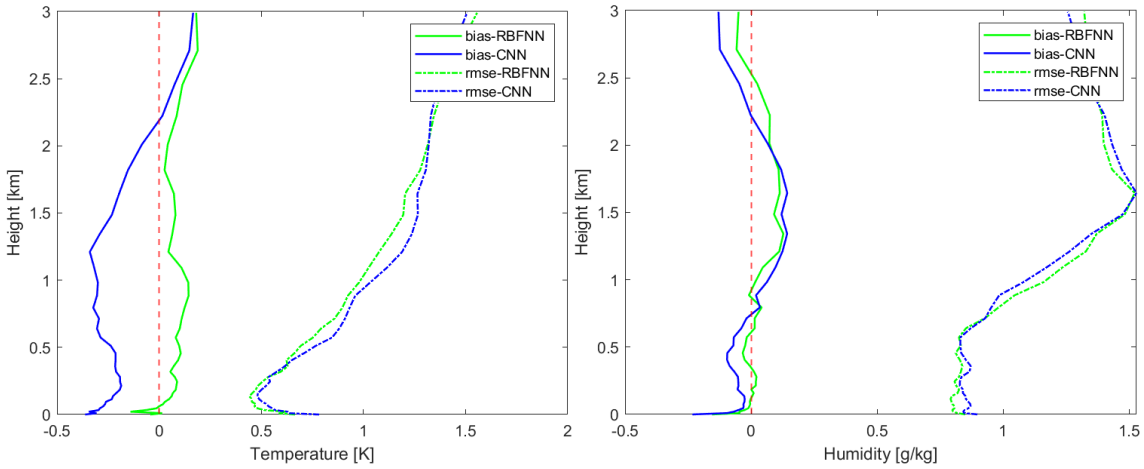


Fig. 13. Comparison of the BIAS and RMSE for RBFNN and CNN. (Left) Results of the temperature profiles and (right) results of the humidity profiles.

interaction between them, thereby reducing the deviation in the retrieval results.

D. Comparison of the Performance of RBFNN and CNN

To further investigate the performance of the inversion performance of RBFNN, the BIAS and RMSE for RBFNN are also compared with the CNN, and the results are shown in Fig. 13. From the figure, it can be seen that, for the retrieval of temperature, the accuracy of RBFNN is slightly higher than CNN, especially for the vertical distribution of BIAS. For the humidity, the performance of RBFNN is similar to that of CNN, but the retrieval accuracy of RBFNN is slightly higher in the low atmosphere (below 500 m).

V. Analysis of Computational Efficiency

To further evaluate the computational efficiency of RBFNN, a comparative analysis of the time consumption among RBFNN, the physical iterative technique, and CNN was conducted. All retrieval processes were performed on a ThinkPad X1 Carbon G9 (CPU: Intel i7, 2.8 GHz; Memory: 16 GB), with the results shown in Table VI.

TABLE VI
Comparison of the Time Consumption of the Three Methods

TIME	Phy	CNN	RBFNN
02/07	19.15 min	3.1 s	0.10 s
04/11	19.73 min	3.4 s	0.12 s
06/06	18.62 min	3.1 s	0.09 s
08/24	19.37 min	3.2 s	0.11 s
10/25	19.29 min	3.3 s	0.09 s
11/21	19.06 min	3.1 s	0.09 s
12/16	19.19 min	3.3 s	0.08 s

It can be found that the computational efficiency of RBFNN is notably higher than the physical iterative method. Specifically, the physical iterative process necessitates approximately 19 min per retrieval, whereas the computation time of our approach is roughly 0.1 s. Compared with CNN, it can be

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:53:08 UTC from IEEE Xplore. Restrictions apply.

found that the computational efficiency of RBFNN is higher than that of CNN on the whole. The reason is that the structure of RBFNN is much simpler than that of CNN, and the computational burden is also smaller.

VI. Conclusion

The ground-based infrared hyperspectral radiometers are an important tool for remote sensing of temperature and humidity profiles in the lower atmosphere. To achieve accurate and efficient retrieval of temperature and humidity profiles, an integrated inversion algorithm is established by using RBFNNs, considering the coupling effect of temperature and water vapor. A joint retrieval strategy is also designed. Due to the strong general approximation capabilities and simple structure of RBFNN, the inversion scheme can learn the nonlinear atmospheric radiative transfer processes from observational data directly, and the temperature and humidity profiles can be simultaneously inverted without iterative solutions. To investigate the performance of RBFNN, a systematic comparison with the eigenvector method and the physical retrieval method is conducted, and the performance of joint and nonjoint retrieval strategies for temperature and humidity is also analyzed. The main conclusions are given as follows.

1) The RBFNN can be effectively applied to the retrieval of temperature and humidity profiles. In the near-surface region, temperature BIAS of the RBFNN was mainly concentrated within ± 1 K, and humidity BIAS remained within ± 1 g/kg across all altitude layers.
2) The retrieval performance of RBFNN is better than that of the eigenvector method and the physical retrieval method. Specifically, the RMSE of the temperature profile for RBFNN is 0.8110 K, and the RMSE of the humidity profile is 1.0070 g/kg, which is at least 12% lower than the other two methods. With the increasing of altitude, the RBFNN also shows higher inversion accuracy in temperature and humidity profiles retrieval.
3) The RBFNN combining temperature and humidity channels demonstrated higher retrieval accuracy, validating the effectiveness of the joint retrieval strategy. The correlation coefficient R^2 for temperature retrieval is improved from 0.9893 to 0.9913, while the correlation coefficient R^2 for humidity retrieval is improved from 0.9424 to 0.9461. This indicates that joint retrieval can more effectively capture the interactions between temperature and humidity, thereby improving the retrieval results.

Due to the limitations of the remote sensing mechanism, ground-based infrared retrievals were mainly carried out under clear-sky conditions. In the next step, we will combine the observations of MWRs and millimeter wave cloud radar to discuss the retrieval of the atmospheric profile under cloudy conditions. Besides, similar to other machine learning-based inversion models, the performance of our inversion model strongly depends on the completeness of the training dataset. Currently, we have only collected long-term time-series data from the SGP site. Therefore, this model may not be fully applicable to other regions with different climatic conditions. In these cases, to ensure the model's reliability and accuracy for other regions with different climates, it requires reestablishing the training database and conducting region-specific training.

Acknowledgment

The authors would like to thank the observation data provided by U.S. Department of Energy's Atmospheric Radiation Measurement (ARM) User Facility. They also thank the anonymous reviewers for taking their valuable time to help them improve the quality of this article.

References

[1] E. H. Shiguemori, H. F. De Campos Velho, J. D. S. Da Silva, and J. C. Carvalho, "Neural network based models in the inversion of temperature vertical profiles from radiation data," *Inverse Problems Sci. Eng.*, vol. 14, no. 5, pp. 543–556, Jul. 2006.

[2] V. Wulfmeyer et al., "A review of the remote sensing of lower tropospheric thermodynamic profiles and its indispensable role for the understanding and the simulation of water and energy cycles," *Rev. Geophys.*, vol. 53, no. 3, pp. 819–895, Sep. 2015.

[3] U. Löhnert, D. D. Turner, and S. Crewell, "Ground-based temperature and humidity profiling using spectral infrared and microwave observations. Part I: Simulated retrieval performance in clear-sky conditions," *J. Appl. Meteorol. Climatol.*, vol. 48, no. 5, pp. 1017–1032, May 2009.

[4] U. Löhnert and O. Maier, "Operational profiling of temperature using ground-based microwave radiometry at payerne: Prospects and challenges," *Atmos. Meas. Techn.*, vol. 5, no. 5, pp. 1121–1134, May 2012.

[5] G. Massaro, I. Stiperski, B. Pospichal, and M. W. Rotach, "Accuracy of retrieving temperature and humidity profiles by ground-based microwave radiometry in truly complex terrain," *Atmos. Meas. Techn.*, vol. 8, no. 8, pp. 3355–3367, Aug. 2015.

[6] D. K. Zhou, W. L. Smith, X. Liu, A. M. Larar, S. A. Mango, and H.-L. Huang, "Physically retrieving cloud and thermodynamic parameters from ultraspectral IR measurements," *J. Atmos. Sci.*, vol. 64, no. 3, pp. 969–982, Mar. 2007.

[7] M. G. Divakarla et al., "Validation of atmospheric infrared sounder temperature and water vapor retrievals with matched radiosonde measurements and forecasts," *J. Geophys. Res., Atmos.*, vol. 111, no. D9, pp. 1–20, Apr. 2006.

[8] P. Martinet et al., "Combining ground-based microwave radiometer and the AROME convective scale model through 1DVAR retrievals in complex terrain: An Alpine valley case study," *Atmos. Meas. Techn.*, vol. 10, no. 9, pp. 3385–3402, Sep. 2017.

[9] D. D. Turner and U. Löhnert, "Ground-based temperature and humidity profiling: Combining active and passive remote sensors," *Atmos. Meas. Techn.*, vol. 14, no. 4, pp. 3033–3048, Apr. 2021.

[10] A. Foth and B. Pospichal, "Optimal estimation of water vapour profiles using a combination of Raman LiDAR and microwave radiometer," *Atmos. Meas. Techn.*, vol. 10, no. 9, pp. 3325–3344, Sep. 2017.

[11] H. Wei, T. Gao, and L. Lei, "Review of remote sensing retrieval of temperature and water vapor profiles using ground-based infrared hyperspectral data," *Infrared*, vol. 36, no. 11, pp. 1–6, 2015.

[12] R. O. Knuteson et al., "Atmospheric emitted radiance Interferometer. Part I: Instrument design," *J. Atmos. Ocean. Technol.*, vol. 21, no. 12, pp. 1763–1776, Dec. 2004.

[13] J. Ye, L. Liu, Q. Wang, S. Hu, and S. Li, "A novel machine learning algorithm for planetary boundary layer height estimation using AERI measurement data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[14] W. Yang, L. Liu, W. Deng, W. Huang, J. Ye, and S. Hu, "Deep retrieval architecture of temperature and humidity profiles from ground-based infrared hyperspectral spectrometer," *Remote Sens.*, vol. 15, no. 9, p. 2320, Apr. 2023.

[15] W. L. Smith, E. Weisz, S. V. Kireev, D. K. Zhou, Z. Li, and E. E. Borbas, "Dual-regression retrieval algorithm for real-time processing of satellite ultraspectral radiances," *J. Appl. Meteorol. Climatol.*, vol. 51, no. 8, pp. 1455–1476, Aug. 2012.

[16] J. Zhang, Z. Li, J. Li, and J. Li, "Ensemble retrieval of atmospheric temperature profiles from AIRS," *Adv. Atmos. Sci.*, vol. 31, no. 3, pp. 559–569, Apr. 2014.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:53:08 UTC from IEEE Xplore. Restrictions apply.

[17] L. Wantong, H. Wei, J. Ming, and S. Jing, "Impact of additional surface meteorological elements onthermodynamic profiles retrieved by ground-based hyper-spectralspectrum based on eigenvector regression algorithm," *Meteorol. Sci. Technol.*, vol. 48, no. 1, pp. 31–39, 2020.

[18] W. L. Smith, W. F. Feltz, R. O. Knuteson, H. E. Revercomb, H. M. Woolf, and H. B. Howell, "The retrieval of planetary boundary layer structure using ground-based infrared spectral radiance measurements," *J. Atmos. Ocean. Technol.*, vol. 16, no. 3, pp. 323–333, Mar. 1999.

[19] D. D. Turner and U. Löhnert, "Information content and uncertainties in thermodynamic profiles and liquid cloud properties retrieved from the ground-based atmospheric emitted radiance interferometer (AERI)," *J. Appl. Meteorol. Climatol.*, vol. 53, no. 3, pp. 752–771, Mar. 2014.

[20] H. Wei, L. Liu, G. Tai-Chang, S. Li, and S. Hu, "The application of the L-curve method in the retrieval of temperature profiles using ground-based hyper-spectral infrared radiance," *Spectrosc. Spect. Anal.*, vol. 36, no. 11, pp. 3620–3624, May 2016.

[21] H. Zhao, X. Ma, G. Jia, Z. Mi, and H. Ji, "Synergistic retrieval of temperature and humidity profiles from space-based and ground-based infrared sounders using an optimal estimation method," *Remote Sens.*, vol. 14, no. 20, p. 5256, Oct. 2022.

[22] Q. Xue, L. Guan, and X. Shi, "One-dimensional variational retrieval of temperature and humidity profiles from the FY4A GIIRS," *Adv. Atmos. Sci.*, vol. 39, no. 3, pp. 471–486, Mar. 2022.

[23] W. L. Smith and H. M. Woolf, "The use of eigenvectors of statistical covariance matrices for interpreting satellite sounding radiometer observations," *J. Atmos. Sci.*, vol. 33, no. 7, pp. 1127–1140, Jul. 1976.

[24] T. Guo, Y. Yang, F. Luo, C. Fu, and L. Zhang, "D^{3}Former: Dual-decoder dual-transformer reconstruction network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5528215.

[25] D. Wang et al., "HyperSIGMA: Hyperspectral intelligence comprehension foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 8, pp. 6427–6444, Aug. 2025.

[26] D. Szwarcman et al., "Prithvi-EO-2.0: A versatile multitemporal foundation model for Earth observation applications," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 4400120, doi: 10.1109/TGRS.2025.3642610.

[27] A. V. Polyakov, Y. M. Timofeev, and Y. A. Virolainen, "Using artificial neural networks in the temperature and humidity sounding of the atmosphere," *Izvestiya, Atmos. Ocean. Phys.*, vol. 50, no. 3, pp. 330–336, Jun. 2014.

[28] S. Sambatti, J. Anochi, E. Pacheco da Luz, E. Shiguemori, A. Carvalho, and H. C. Velho, "Automatic configuration for neural network applied to atmospheric temperature profile identification," in *Proc. Int. Conf. Eng. Optim.*, 2012, pp. 1–9.

[29] D. Malmgren-Hansen, V. Laparra, A. A. Nielsen, and G. Camps-Valls, "Statistical retrieval of atmospheric profiles with deep convolutional neural networks," *ISPRS J. Photogramm. Remote Sens.*, vol. 158, pp. 231–240, Dec. 2019.

[30] Y. Xiao et al., "Integrated retrieval of the temperature and humidity profiles of atmospheric boundary layer by combining ground-based infrared hyperspectral interferometers and microwave radiometers," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2024, Art. no. 4101213, doi: 10.1109/TGRS.2024.3520231.

[31] X. Cai et al., "Temperature and humidity profile retrieval from FY4-GIIRS hyperspectral data using artificial neural networks," *Remote Sens.*, vol. 12, no. 11, p. 1872, Jun. 2020.

[32] G. Zheng et al., "Using artificial neural network ensembles with crogging resampling technique to retrieve sea surface temperature from HY-2A scanning microwave radiometer data," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 985–1000, Feb. 2019.

[33] X. Yan, C. Liang, Y. Jiang, N. Luo, Z. Zang, and Z. Li, "A deep learning approach to improve the retrieval of temperature and humidity profiles from a ground-based microwave radiometer," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8427–8437, Dec. 2020.

[34] J. Yang, C. Wu, B. Du, and L. Zhang, "Enhanced multiscale feature fusion network for HSI classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10328–10347, Dec. 2021.

[35] C. Fu, T. Zhou, T. Guo, Q. Zhu, F. Luo, and B. Du, "CNN-transformer and channel-spatial attention based network for hyperspectral image classification with few samples," *Neural Netw.*, vol. 186, Jun. 2025, Art. no. 107283.

[36] J. Zhang et al., "Multiscale spatial–spectral CNN-transformer network for hyperspectral image super-resolution," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 12116–12132, 2025.

[37] D. S. Broomhead and D. Lowe, "Multivariable functional interpolation and adaptive networks," *Complex Syst.*, vol. 2, pp. 321–355, Aug. 1988.

[38] H.-G. Han, M.-L. Ma, and J.-F. Qiao, "Accelerated gradient algorithm for RBF neural network," *Neurocomputing*, vol. 441, pp. 237–247, Jun. 2021.

[39] J. Qiao and H. Han, "Identification and modeling of nonlinear dynamical systems using a novel self-organizing RBF-based approach," *Automatica*, vol. 48, no. 8, pp. 1729–1734, 2012.

[40] X. Shi, Y. Cheng, C. Yin, X. Huang, and S. Zhong, "Design of adaptive backstepping dynamic surface control method with RBF neural network for uncertain nonlinear system," *Neurocomputing*, vol. 330, pp. 490–503, Feb. 2018.

[41] X. Ben, W. Meng, R. Yan, and K. Wang, "Kernel coupled distance metric learning for gait recognition and face recognition," *Neurocomputing*, vol. 120, pp. 577–589, Nov. 2013.

[42] Y. Ma, L. Ma, and J. Ma, "RBF neural network based fault diagnosis for the thermodynamic system of a thermal power generating unit," in *Proc. Int. Conf. Mach. Learn. Cybern.*, Aug. 2005, pp. 4738–4743.

[43] C. Arteaga and I. Marrero, "Universal approximation by radial basis function networks of Delsarte translates," *Neural Netw.*, vol. 46, pp. 299–305, Oct. 2013.

[44] W. F. Feltz, W. L. Smith, R. O. Knuteson, H. E. Revercomb, H. M. Woolf, and H. B. Howell, "Meteorological applications of temperature and water vapor retrievals from the ground-based atmospheric emitted radiance interferometer (AERI)," *J. Appl. Meteorol.*, vol. 37, no. 9, pp. 857–875, Sep. 1998.

[45] K.-L. Du and M. N. S. Swamy, "Radial basis function networks," in *Neural Networks and Statistical Learning*. London, U.K.: Springer, 2013.

[46] M. J. D. Powell, "Radial basis functions for multivariable interpolation: A review," in *Algorithms for Approximation*. Oxford, U.K.: Clarendon, Jan. 1987, pp. 143–167.

[47] Y. Wu, H. Wang, B. Zhang, and K.-L. Du, "Using radial basis function networks for function approximation and classification," *ISRN Appl. Math.*, vol. 2012, Mar. 2012, Art. no. 324194.

[48] W. Huang et al., "Retrieval of temperature and humidity profiles from ground-based high-resolution infrared observations using an adaptive fast iterative algorithm," *Atmos. Meas. Techn.*, vol. 16, no. 17, pp. 4101–4114, Sep. 2023.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:53:08 UTC from IEEE Xplore. Restrictions apply.